

Sentinel

A Fully Procedural Game Engine
and Open-World Military Shooter

*Nothing is loaded from a model, image or sound file:
everything is generated from code and a seed.*

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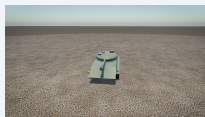
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Outline

1. Problem, objectives, architecture
2. Procedural shapes and textures
3. Transformations: creating and combining objects
4. Illumination models and light sources
5. Rendering pipeline: shadows, ambient, tone map
6. Results, verification, a bug fixed
7. Video tour of the project



Six illumination models



1. Problem, objectives and architecture

Can a complete, playable 3D game be made by code alone on a student laptop?

TWO HARD LIMITS

No model, image or precomputed file; OpenGL 4.1, no compute shaders.

OBJECTIVES

Layered engine; all content from code; modern renderer; a playable game (vehicles, ladders, stealth, five-objective mission); tested, 16.7 ms budget.

Game: mission, AI, vehicles

Render: deferred, lights, sky

World: collision, terrain

Procedural: noise, textures

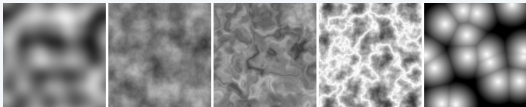
GPU: shaders, buffers

Platform: window, input



A lower layer never knows a higher one.

2. Procedural shapes and textures



gradient, fractal, warped, ridged, cellular noise

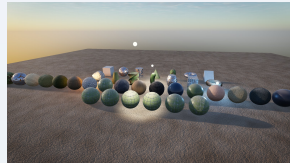
FRACTAL NOISE (TILES EXACTLY)

$$F(\mathbf{x}) = \frac{\sum_{i=0}^{n-1} 2^{-i} v(2^i \mathbf{x})}{\sum_{i=0}^{n-1} 2^{-i}}$$

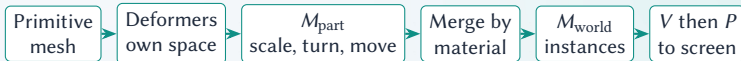
VERTEX PUSH BY A NOISE DEFORMER

$$\mathbf{p}' = \mathbf{p} + A v(f\mathbf{p}) \hat{n}, \quad \|\mathbf{p}' - \mathbf{p}\| \leq A, \quad v \in [-1, 1]$$

- 18 recipes $\rightarrow$ 22 materials, baked once at 1024^2
- **Scharr** filter turns height into bumps; crevices darken
- Objects: tables of primitives and noise deformers
- Soldiers: skeletons with two-bone IK legs



3. Transformations: creating and combining objects



ONE PART: SCALE, TURN, THEN MOVE

$$M_{\text{part}} = T(\mathbf{t}) R_z(\gamma) R_y(\beta) R_x(\alpha) S(\mathbf{s})$$

Right to left: stretched, turned, moved last.

$$(1, 0, 0) \xrightarrow{S(2,1,1)} (2, 0, 0) \xrightarrow{R_y(90^\circ)} (0, 0, -2) \xrightarrow{T(5,0,0)} (5, 0, -2)$$

NORMALS USE THE INVERSE TRANSPOSE

$$\hat{n}' = \frac{(L^{-1})^\top \hat{n}}{\|(L^{-1})^\top \hat{n}\|}, \quad L = \text{upper-left } 3 \times 3 \text{ of } M$$

$$S = \text{diag}(2, 1, 1), \hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0): \text{correct } (0.447, 0.894, 0), \\ \text{naive } (0.894, 0.447, 0)$$

ROTATION ABOUT y

$$R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

HIERARCHY: PRODUCTS DOWN THE CHAIN

$$M_{\text{gun}} = M_{\text{hull}} T(\mathbf{c}_t) R_y(\theta_t) T(\mathbf{c}_g) R_x(-\theta_g)$$

WORLD TO SCREEN

$$\mathbf{p}_{\text{clip}} = P V M_{\text{world}} M_{\text{part}} \mathbf{p}$$

4. Illumination models and light sources

DIRECT LIGHT + AMBIENT + EMISSION

F: MODEL ATT: FALLOFF SH: SHADOW N_L : LIGHTS

$$L_o = \sum_{j=1}^{N_L} f(l_j, \mathbf{v}) L_j \text{ att}_j \text{ sh}_j \max(0, \mathbf{n} \cdot \mathbf{l}_j) + \text{ambient} \cdot \text{AO} + \mathbf{e}$$

SIX ILLUMINATION MODELS

Lambert

$$f = \frac{c_d}{\pi}$$

Phong

$$f = \frac{c_d (1 - F_0)}{\pi} + F_0 \frac{m + 2}{2\pi} (\mathbf{r} \cdot \mathbf{v})^m$$

Blinn-Phong

$$f = \frac{c_d (1 - F_0)}{\pi} + F_0 \frac{m + 8}{8\pi} (\mathbf{n} \cdot \mathbf{h})^m$$

Oren-Nayar

$$f = \frac{c_d}{\pi} (A + B \max(0, \cos \Delta\varphi) \sin \alpha \tan \beta)$$

Cook-

Torrance

$$f = \frac{(1 - F_l)(1 - F_v) c_d}{\pi} + D V F$$

Subsurface

$$\langle \mathbf{n} \cdot \mathbf{l} \rangle_w = \text{clamp}\left(\frac{\mathbf{n} \cdot \mathbf{l} + w}{1 + w}, 0, 1\right), \quad w = 0.5$$

LIGHT SOURCES

$$\text{att}(d) = \begin{cases} \frac{(1 - (d/R)^4)^2}{d^2 + 1}, & 0 \leq d \leq R \\ 0, & d > R \end{cases}$$

DEFINITIONS

$$m = \max\left(\frac{2}{\rho^2 + 10^{-4}} - 2, 1\right), \quad \mathbf{r} = 2(\mathbf{n} \cdot \mathbf{l}) \mathbf{n} - \mathbf{l}, \quad \mathbf{h} = \frac{\mathbf{l} + \mathbf{v}}{\|\mathbf{l} + \mathbf{v}\|}$$

$$A = 1 - \frac{0.5 \sigma^2}{\sigma^2 + 0.33}, \quad B = \frac{0.45 \sigma^2}{\sigma^2 + 0.09}, \quad \sigma = \rho$$

$$D = \frac{\alpha^2}{\pi ((\mathbf{n} \cdot \mathbf{h})^2 (\alpha^2 - 1) + 1)^2}, \quad F = F_0 + (1 - F_0) (1 - \mathbf{v} \cdot \mathbf{h})^5, \quad \alpha = \rho^2$$

5. Shadows, ambient light and tone mapping



Deferred shading: store colour, normal, shininess and depth first; light each pixel once.

CASCADED SHADOW MAPS

$$z_i = \lambda n \left(\frac{r}{n} \right)^{i/N} + (1 - \lambda) \left(n + (r - n) \frac{i}{N} \right), \quad i = 1, \dots, N$$

$$\lambda = 0.75, N = 3, n = 0.1, r = 100$$

cuts $z_{1,2,3} = 9.1, 24.2, 100$ m; normal-offset lookup,
8 rotated PCF taps.

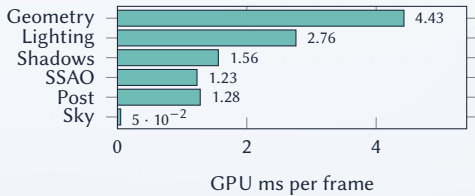
FOG AND TONE MAP

$$C = T C_{\text{surface}} + (1 - T) C_{\text{haze}}, \quad T = e^{-kd}$$
$$C_{\text{out}} = \frac{x(2.51x + 0.03)}{x(2.43x + 0.59) + 0.14}, \quad x = \text{exposure} \times \text{radiance}$$



Dusk: sun, sky, haze, shadows
Ambient = sky diffuse + split-sum
specular, times SSAO.

6. Results, verification and a bug fixed



11.3 ms of the 16.7 ms budget at 1080p

Unit tests (13 packages)	281
TextureCheck (exact tiling)	8 318
RenderCheck (lighting probes)	318 567
Gpu, Window, Audio checks	30, 141, 79
Playthroughs, 5/5 objectives	2

Bug: a vertical line at night. Fog sampled the whole sky, so one horizon star streaked a column; fog now uses the smooth atmosphere only.



before



after

LIMITS AND FUTURE WORK

Blocky models, no true global illumination, lumpy foliage. Next: leaf cards, screen-space GI, point-light shadows, multiplayer.

I 7. Video tour of the project

References

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Thank you

Questions are welcome
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